Miscellaneous Problems

1. 0.906 2. 0.382 3. 0.95 4. 0.667

5. 0.905 6. Y=0.653X+21.71; Marks in B = 55.67

7. Y=0.576X+2.332; Y =5.788 8. Y=0.867X+7.913

9. Y=–0.652X+63.945, Y=10.481≈10 10. b_{yx}=1.422, Y = 141.67

10. OPERATIONS RESEARCH

Exercise: 10.1

1. Maximize $Z = 5x_1 + 3x_2$ subject to constraints $2x_1 + x_2 \leq 1000$; $x_1 \leq 400; x_2 \leq 700$ and $x_1, x_2 \geq 0$.

2. Maximize $Z = 30x_1 + 40x_2$ subject to constraints $60x_1 + 120x_2 \leq 12000$ $8x_1 + 5x_2 \leq 600; 3x_1 + 4x_2 \leq 500$ and $x_1, x_2 \geq 0$.

3. Maximize Z=$100x_1$+$150x_2$ subject to constraints $0.8x_1 + 1.2x_2 \leq 720; x_1 \leq 600;$ $x_2 \leq 400$ and $x_1, x_2 \geq 0$.

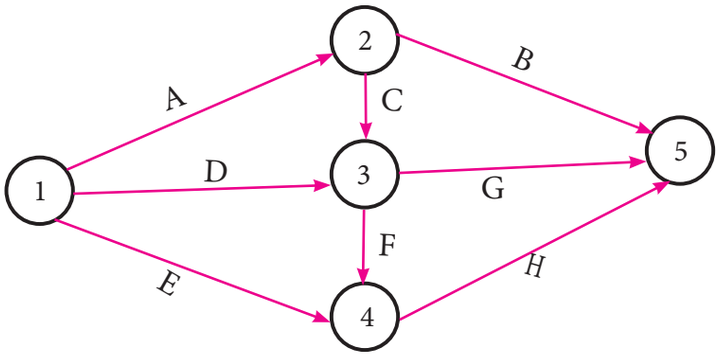
4.(i) $x_1 = 4$; $x_2 = 9$ and $Z_{max} = 96$ (ii) $x_1 = 8$; $x_2 = 12$ and $Z_{max} = 392$

(iii) $x_1 = 1$; $x_2 = 5$ and $Z_{min} = 13$ (iv) $x_1 = 2$; $x_2 = 3$ and $Z_{max} = 230$

(v) $x_1 = 3$; $x_2 = 9$ and $Z_{max} = 330$ (vi) $x_1 = 4$; $x_2 = 2$ and $Z_{min} = 160$

Exercise: 10.2

1.



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